

Bank Card Transaction Analysis

Overview

The project focuses on analyzing card payment transactions of a bank in order to identify meaningful spending patterns across customer groups, merchant categories, geographical regions, card types and time periods.

Customers use payment cards linked to their bank accounts to make payments at different merchants, generating the transaction data used for the analysis.

The analysis is intended to provide the bank with a clearer understanding of where and when transaction volume is generated and which customer and card segments contribute most to it.

Objective

The objective is to provide bank management and card product or marketing teams with insights from historical card transaction data that can support business decisions related to card products and customer targeting.

The system should help answer questions such as:

- Which customer groups generate the highest transaction volume in specific merchant categories?
- In which regions and merchant categories is card spending highest?
- How do transaction value and number of transactions change across months, quarters, seasons or years?
- How do transaction value and number of transactions differ between card types?

These insights can support decisions such as designing card offers and reward programs, targeting relevant customer groups, and identifying attractive merchant categories or regions for promotions and partnerships.

Scope / Constraints

The project focuses exclusively on **card payment transactions**.

Included in the project are:

- bank customers,
- customer accounts,
- payment cards,

- card transactions,
- merchants,
- merchant categories,
- transaction locations and time periods.

The following banking areas are outside the scope of the project:

- loans and mortgages,
- bank transfers,
- investments,
- cash withdrawals,
- fraud detection,
- credit risk,
- online banking,
- and other banking products not directly related to card payments.
- A merchant represents an individual merchant location identified by its MerchantID. Merchant chains or brands are not modeled separately within the scope of this project.

Business Scenario

A bank provides its customers with bank accounts and payment cards.

Customers use these cards to make payments at different merchants. Each transaction contains information such as the transaction amount, date and time, the card used and the merchant where the payment took place.

Merchants operate in different business categories, such as groceries, restaurants, electronics or clothing, and may be located in different cities or countries.

The general business process can therefore be described as:

Customer → Account → Card → Transaction → Merchant